PetCare AI Setup Guide

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13 OCTOBER 2026

This guide sets up the Windows dashboard and Raspberry Pi service so they can exchange live sensor readings on the same local network. Follow the order below: install the dashboard, start its receiver, then start the Pi program.

Keep your Telegram bot token private. The screenshots supplied with the setup request show a token. Revoke that exposed token with BotFather and use a replacement before sharing screenshots or configuring a friend's copy. Do not put tokens in this guide.

1. Extract and start the Windows GUI

1. Extract **PetCareAI_Current_2026-10-13.zip** fully. Open PowerShell in the extracted folder.

2. Install Python 3.11 if needed. Create the GUI environment and install the tested packages:

```
cd GUI
py -3.11 -m venv .venv-local
.\.venv-local\Scripts\python.exe -m pip install -r requirements-tested.txt
```

3. Start the application:

```
.\.venv-local\Scripts\python.exe main.py
```

After first setup, double-click **GUI/Launch PetCare AI.vbs**. Demo sign-in: **admin** / **pet123**.

2. Start the Live Pi receiver on Windows

4. In PowerShell run **ipconfig**. Find the IPv4 Address under the Wi-Fi adapter currently connected to the same network as the Pi. It can change, so check it after a network change.

5. In PetCare AI, open **Sensor Hub**, choose **Live Pi**, open **Connect Pi / Gesture access**, keep Listen address **0.0.0.0** and Port **5005**, then click **Start receiver**.

6. If Windows Firewall asks, allow the GUI's **python.exe** on the trusted private network.

Keep the GUI and receiver running while the Pi connects.

3. Wire and start the Raspberry Pi

Copy the package's **raspberry_pi** folder to the Pi. Keep **predict_movement.py** and **movement_knn_model.joblib** together. The pin numbers below are physical BOARD header numbers, not BCM GPIO numbers.

Input	Default role	TRIG (BOARD)	ECHO (BOARD)
Sensor 1	Food level	13	15
Sensor 2	Sleeping	16	18
Sensor 3	Water level	22	24
Sensor 4	Toilet	29	31
Sensor 5	Gesture KNN	7	11

Before connecting: switch off the Pi before rewiring. Give sensors and Pi a common ground. HC-SR04 ECHO outputs are 5 V; put a voltage divider or level shifter on every ECHO line before it reaches a Pi GPIO. Confirm wiring and avoid pins used by a HAT or other hardware.

Create the Pi environment and install requirements

```
cd ~/Desktop/raspberry_pi
python3 -m venv env --system-site-packages
source env/bin/activate
python -m pip install -r requirements.txt
```

Start the Windows receiver first. Then launch the Pi script, replacing the example with the Windows Wi-Fi IPv4 address from Step 4:

```
python predict_movement.py --gui-host YOUR_LAPTOP_IPV4
```

A successful connection prints messages similar to **[TCP] Connected** and **[CONFIG] Applied**. Stop with **Ctrl+C**. The model was trained with scikit-learn 1.6.1; the Pi requirements pin this version.

Quick no-hardware connection check

To confirm the network and receiver before connecting sensors, run from the Pi folder:

```
python predict_movement.py --gui-host YOUR_LAPTOP_IPV4 --simulate
```

Simulator messages are synthetic and appear in Live Pi history. Stop it with Ctrl+C before starting the hardware service.

4. Set sensor roles and calibrate

1. In **Sensor Hub**, choose the function for Sensors 1-4 and click **Save assignments & send to Pi**. Sensor 5 is reserved for entrance gesture KNN.
2. Default presence thresholds: sleeping below **14 cm**; toilet below **13 cm**. These readings estimate time near a station. They do not prove the pet is asleep or using the toilet.
3. For food and water, open Container calibration. Measure the sensor distance when each container is full and empty, enter both values, and save. Enter container capacity for estimated litres. Use the uniform cross-section option only if container width stays about the same with height.
4. Open Live Sensors and check that fresh distances and timestamps update. A missing echo is shown as unavailable, not as zero.

5. Configure Telegram and verify access

1. In Telegram, open your own bot and press **Start** or send **/start**. Use your own BotFather token and the intended recipient chat ID.
2. In PetCare AI, open **Settings > Connections**. Select **Real**, enter the replacement bot token and chat ID, keep Enable Telegram notifications checked and Remove saved bot token unchecked, then click Save settings.
3. Click **Test saved Telegram connection**. Continue after the test message arrives. If it fails, check the token, chat ID, /start on that bot, and the laptop internet connection.
4. To test OTP, perform the trained gesture at Sensor 5 and wait for **AUTH_GESTURE**. Enter the registered username (demo account: admin), click Request OTP, enter the six-digit Telegram code, then click Verify OTP.
5. Enable the entrance approach-and-stay category under Settings > Notifications to send **APPROACH_STAY** alerts to Telegram.

Troubleshooting

What you see	Check
Connection unavailable	Receiver running; current laptop IP; both devices on reachable network; TCP 5005 allowed in Firewall.
CONFIG not applied	Pi folder writable and GUI receiver running. Restart Pi service after receiver is ready.
No valid echo	BOARD pins, common ground, ECHO divider, sensor power, direction and range. Test one sensor at a time.
OTP delivery failed	Replacement token and correct chat ID; start the bot chat; save Settings; test saved connection.

Privacy: supplied screenshots expose the old token. Revoke it in BotFather and enter a replacement only in the app. Each friend should configure their own token, chat ID, pet profile and container calibration. The Pi-to-GUI lab TCP connection is unencrypted; use a trusted network.

The setup archive contains no personal Telegram credentials or saved database.